Feedback Loops

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One current feedback-loop record, with earlier implementation drafts preserved.

Trace displays/clicks, interpret a nonclick, repair a raw-count claim and limit simulated diversity evidence.

Keep one habit: Current tasks have separate answer fields; earlier questions and saved responses remain labelled review.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your **explanation**.

- Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Recommendation Scores

Similarity by Selected Traits

Missing Data

MAIN LESSON · SECTION 12

W1 · Read display opportunities and clicks

> Current responses are separate from retained implementation answers.

Current event contract

Current simulation: both books start at0 displays/0 clicks; click-count ties chooseA. Scripted A displays click and B displays do not. Compare3rounds of click-count selection with the supplied A,B,A plan. These are fictional events, not reader measurements. Keep exposure and click denominators separate.

Current task and new answer field

- From the supplied click-count-policy log, give A and B displays and clicks after each round and at the end. Explain why B's zero clicks differ from a reported dislike.

My current feedback-loop reasoning and evidence

▼ **Review · Earlier implementation task and saved response**

Why does A win the worked example even though B is **process**ed first?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Interpret a planned nonclick

> Current responses are separate from retained implementation answers.

Current event contract

Current simulation: both books start at0 displays/0 clicks; click-count ties chooseA. Scripted A displays click and B displays do not. Compare3rounds of click-count selection with the supplied A,B,A plan. These are fictional events, not reader measurements. Keep exposure and click denominators separate.

Current task and new answer field

- B was never shown, then a planned trial shows B once with no click. Give the before/after exposure and click counts, identify which click rate is unavailable and limit the inference.

My current feedback-loop reasoning and evidence

▼ **Review · Earlier implementation task and saved response**

State recommend([], 10) and the worked items with limit 8.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair a raw-click preference claim

> Current responses are separate from retained implementation answers.

Current event contract

Current simulation: both books start at0 displays/0 clicks; click-count ties chooseA. Scripted A displays click and B displays do not. Compare3rounds of click-count selection with the supplied A,B,A plan. These are fictional events, not reader measurements. Keep exposure and click denominators separate.

Current task and new answer field

- A report says A must be preferred because its three clicks exceed B’s zero, but only A was displayed. Rewrite the conclusion and propose a predeclared exposure comparison; do not invent future responses.

My current feedback-loop reasoning and evidence

▼ **Review · Earlier implementation task and saved response**

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit the simulated diversity result

> Current responses are separate from retained implementation answers.

Current event contract

Current simulation: both books start at0 displays/0 clicks; click-count ties chooseA. Scripted A displays click and B displays do not. Compare3rounds of click-count selection with the supplied A,B,A plan. These are fictional events, not reader measurements. Keep exposure and click denominators separate.

Current task and new answer field

- The supplied planned exposure run displays two distinct books instead of one. Does this establish higher enjoyment or educational benefit? State what was scripted, what was counted and what further evidence would be needed.

My current feedback-loop reasoning and evidence

▼ **Review · Earlier implementation task and saved response**

Write a one-sentence **explanation** of X winning without calling it objectively best.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Audit the new X/Y/Z log independently

> Keep the first attempt before feedback.

Current independent task and new response

- A fresh fictional log has X displayed4 times with1 click, Y displayed4 times with2 clicks, and Z displayed0 times with0 clicks. Calculate the observed click rates for X and Y; state why Z has no measured click rate. Explain how always selecting the highest raw-click item can shape future evidence. Then a planned trial shows Z once with no click: update Z's counts and rate, and state one limit of that observation.

> **My independent exposure-log reasoning**

Fair-attempt record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Review · Earlier implementation task and saved response**

Independent case: Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Handle no match for Implementing a **recommender**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Implementing a **recommender**. Proposed **explanation**: "A successful run proves the conclusion". Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the **feedback loop**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 24 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

My explanation after practice

Record support used: none, hint or teacher **explanation**.

▼ Review · Earlier implementation transfer

REFERENCE · SECTION 15

Review · Earlier implementation transfer

> This old task is not a copy of the current feedback-log W4.

Earlier task and saved response

- Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

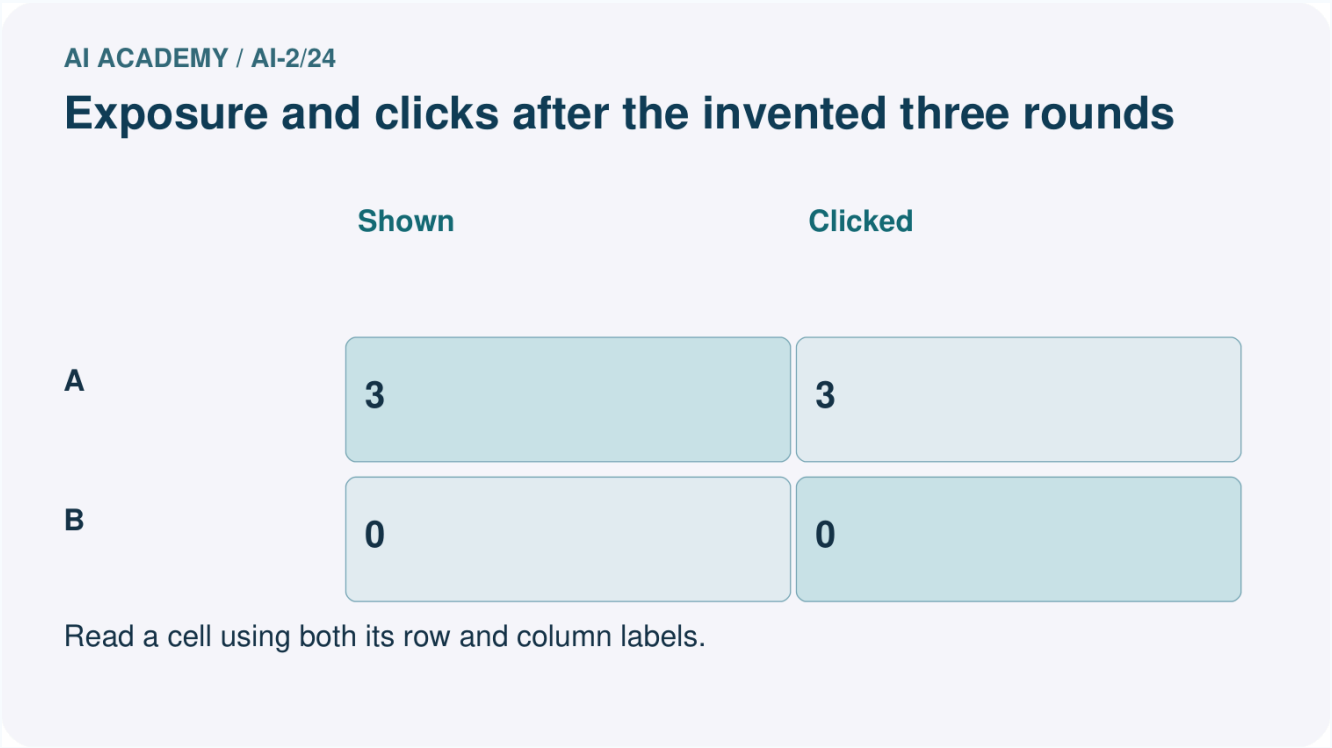
> Keep the explanation and the support you used.

Use this when helpful

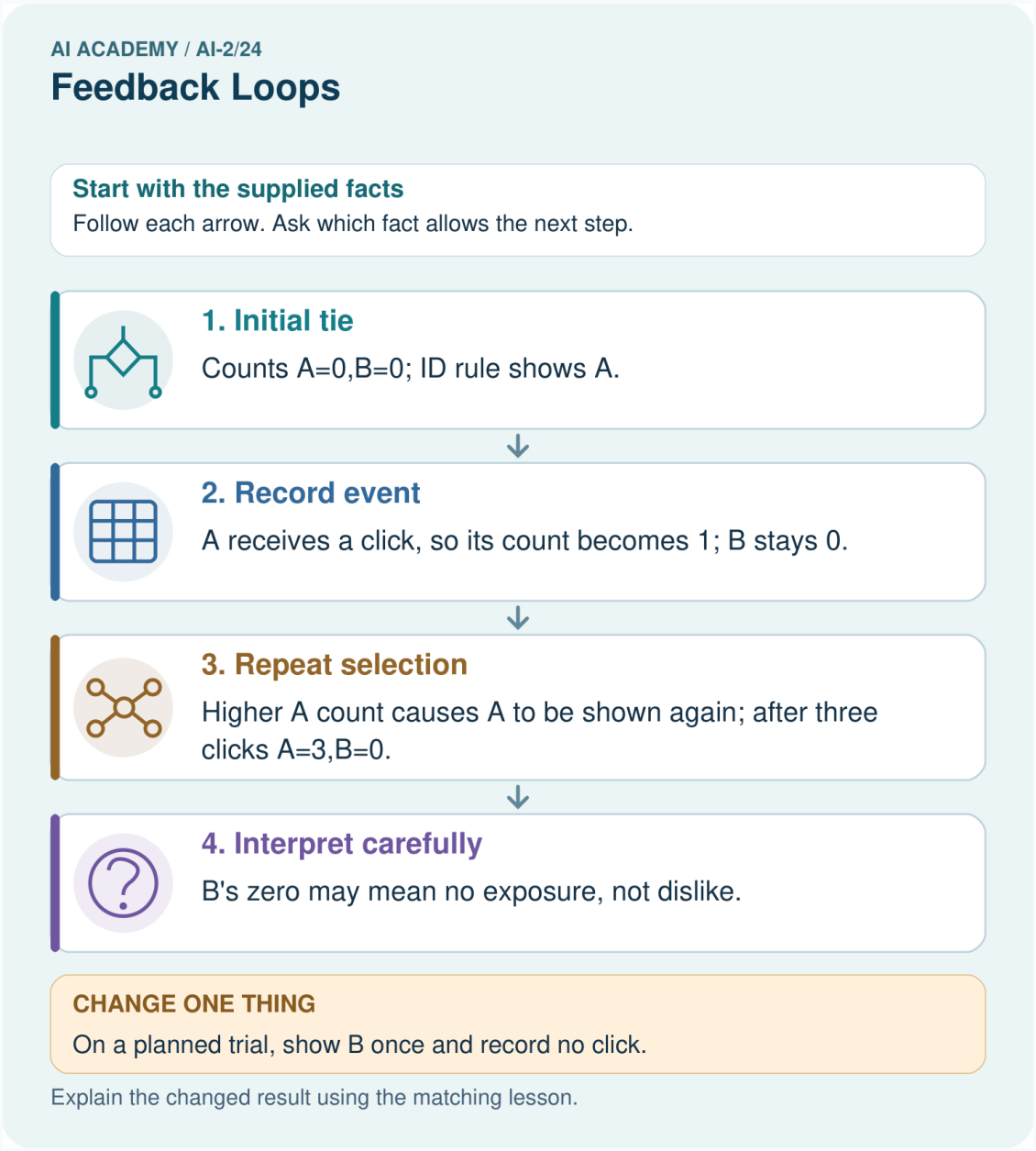
A fictional catalogue repeatedly recommends two books using observed clicks.

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

What extra field would reveal why comparing raw clicks is misleading?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book **Recommender**

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue repeatedly recommends two books using observed clicks.
- Which labelled step matches this detail? B's zero may mean no **exposure**, not dislike.
- 1. Record event
- 2. Repeat selection
- 3. Initial tie
- 4. Interpret carefully
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported **explanation**: Interpret carefully B's zero may mean no **exposure**, not dislike. The selection policy changes which **evidence** is available.

A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Changed-case question

- Changed-case practice: On a planned trial, show B once and record no click. Which conclusion is supported by the supplied facts?
- 1. B gains one **exposure** but still zero clicks.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported **explanation**: B gains one **exposure** but still zero clicks. The **observation** now supports a nonclick on that display, not a permanent dislike.

Repair and save your work

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Feedback Loops: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

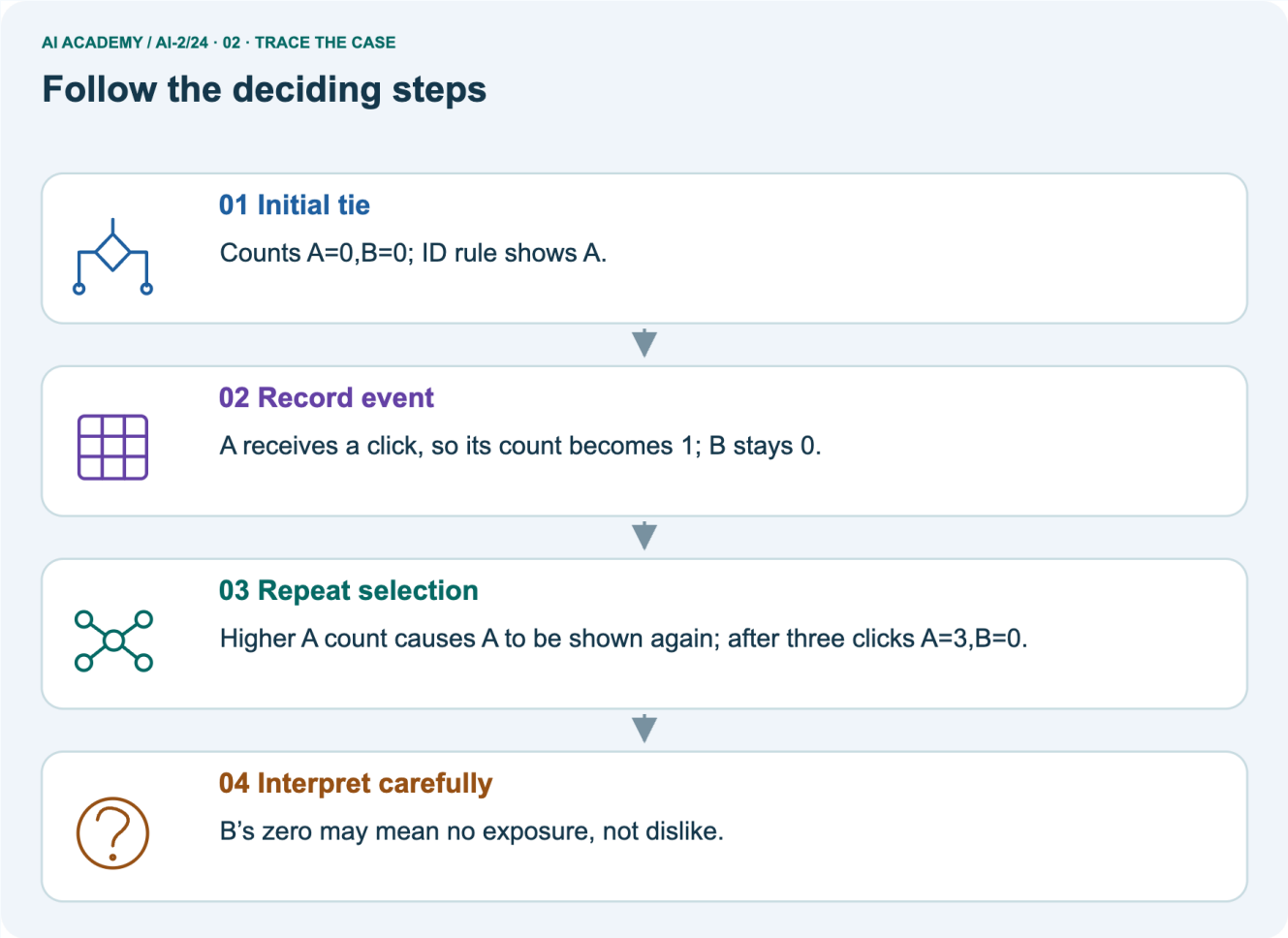
A fictional catalogue repeatedly recommends two books using observed clicks. Supplied facts: Initially each click count is 0. Equal counts choose earlier ID A over B. Only the recommended book is shown.

In an invented three-round record, A is shown and clicked each round; B is never shown.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



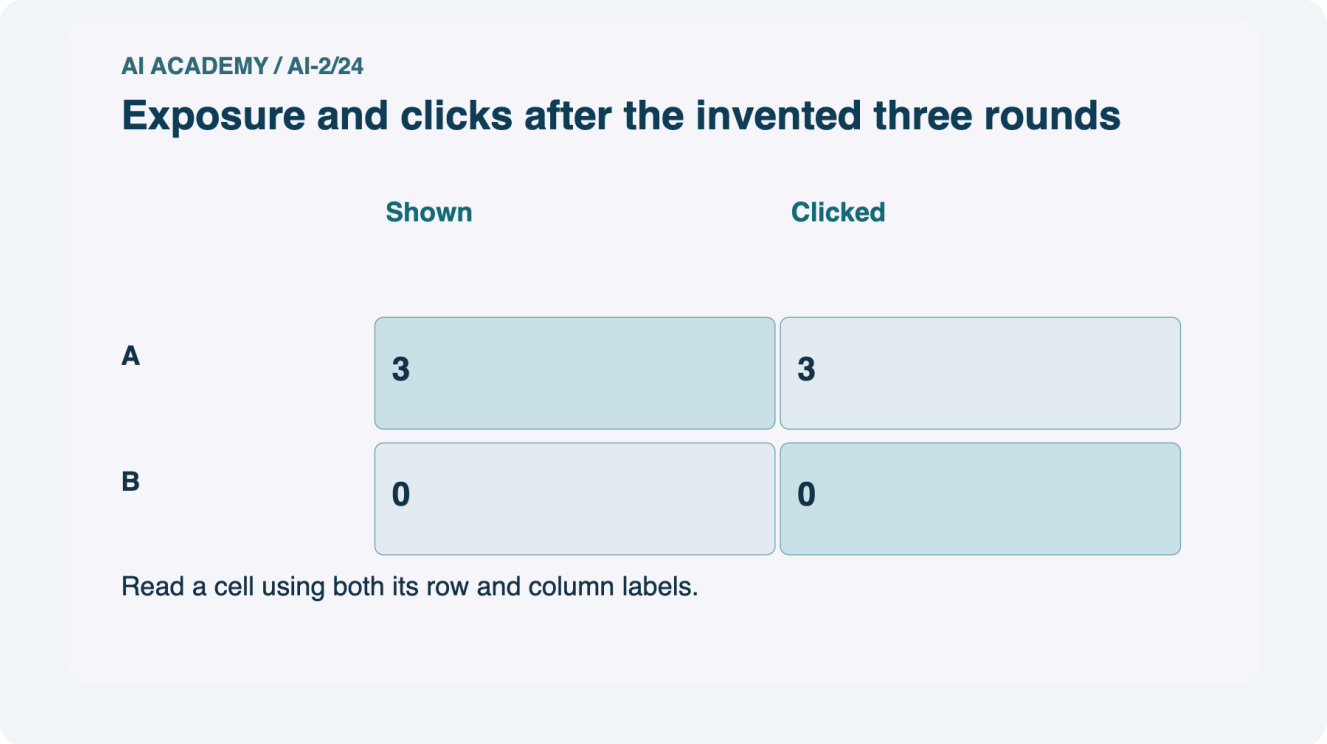
Follow the labelled picture with the matching explanation. Open the vector diagram

Initial tie: Counts A=0,B=0; ID rule shows A. → Record event: A receives a click, so its count becomes 1; B stays 0. → Repeat selection: Higher A count causes A to be shown again; after three clicks A=3,B=0. → Interpret carefully: B's zero may mean no **exposure**, not dislike.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Exposure and clicks after the invented three rounds



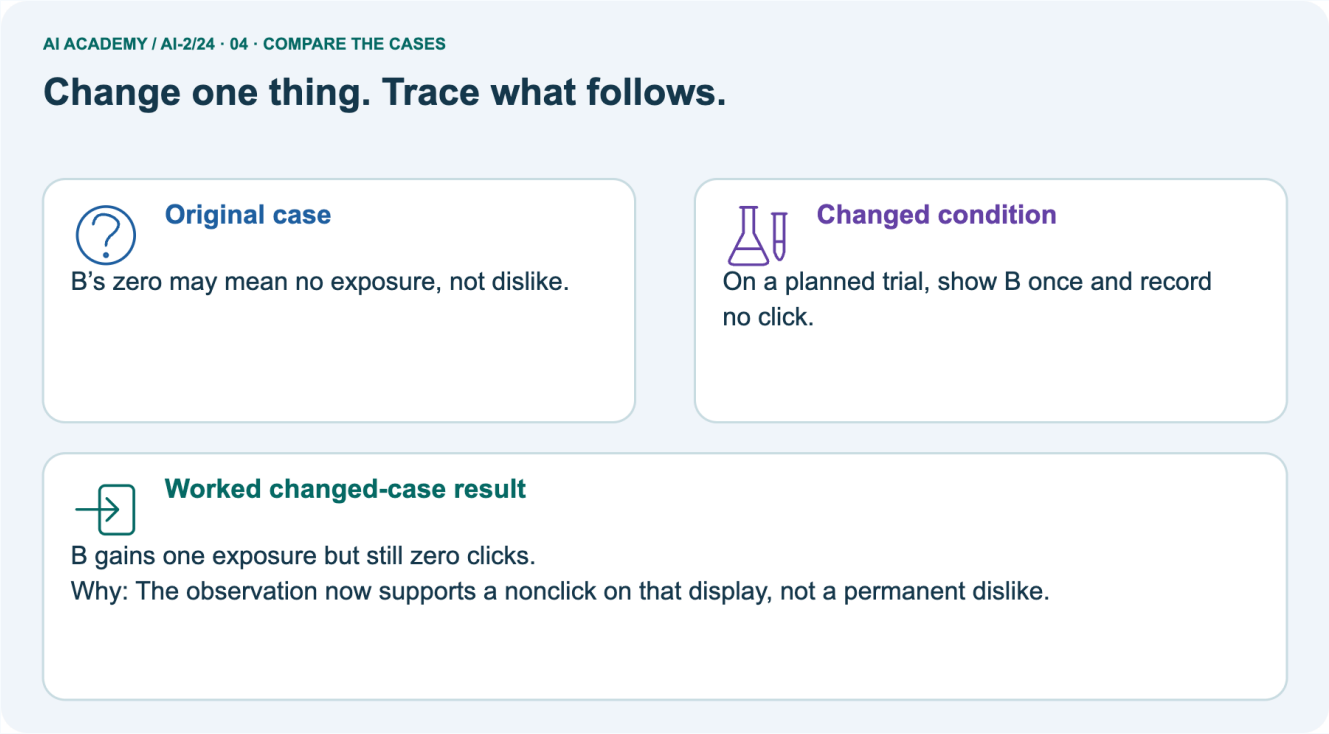
Follow the labelled picture with the matching explanation. Open the vector diagram

Exposure and clicks after the invented three rounds. A: Shown = 3, Clicked = 3. B: Shown = 0, Clicked = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

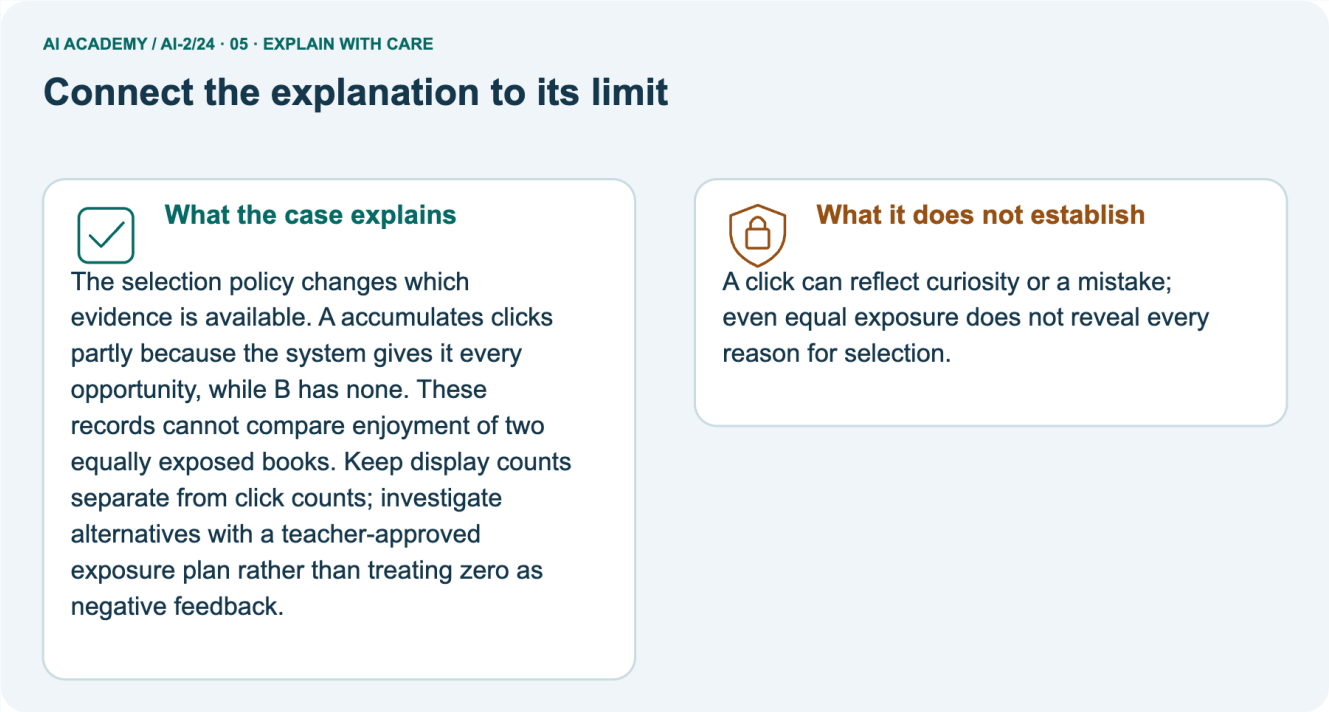
Original case: B's zero may mean no **exposure**, not dislike. Changed condition: On a planned trial, show B once and record no click. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

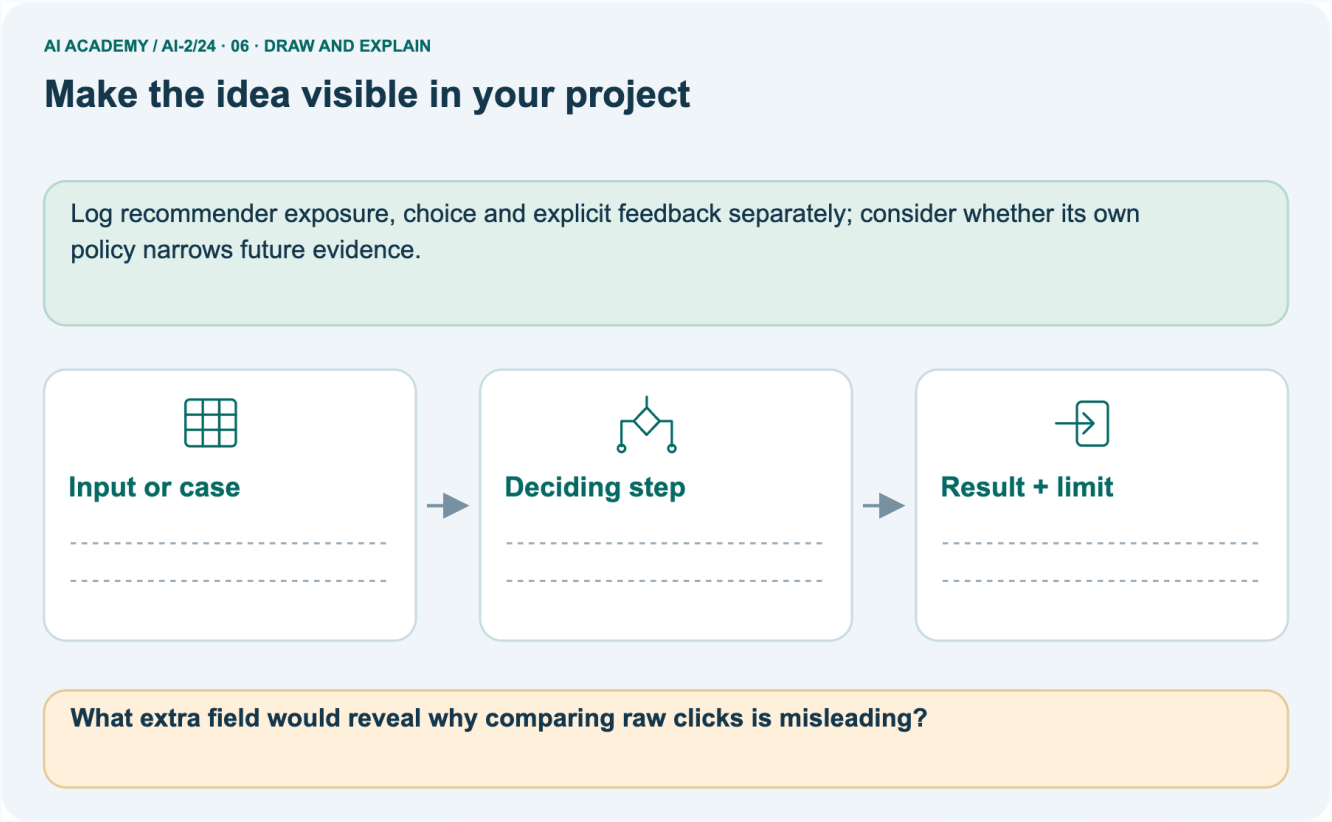
Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Limit: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: What extra field would reveal why comparing raw clicks is misleading?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Feedback Loops: the case

02. Follow the deciding steps

03. Exposure and clicks after the invented three rounds

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.